Second Order Partial Differential Equations

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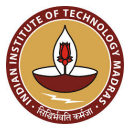


Second Order Linear PDEs

- A general second order linear PDE in two independent variables x, y is

$$a\,u_{xx} + b\,u_{xy} + c\,u_{yy} + d\,u_x + e\,u_y + f\,u = g. \tag{1}$$

- Here a, b, c, d, e, f, g are functions of x and y only.
- a, b, c do not vanish simultaneously at any point of the domain.
- We assume these functions and u have continuous second order partial derivatives.
- Then $u_{xy} = u_{yx}$, so only one mixed derivative appears.
- The equation is linear: u and its derivatives occur only to the first power.
- No products such as $u\,u_x$ or $u_{xx}u_{yy}$ occur.



Principal Part

- The terms containing second derivatives,

$$a\,u_{xx} + b\,u_{xy} + c\,u_{yy},$$

  form the principal part of the PDE.
- The remaining terms $d\,u_x + e\,u_y + f\,u$ are the lower order terms.
- The principal part decides the qualitative nature of the equation.
- We will often write (1) as

$$a\,u_{xx} + b\,u_{xy} + c\,u_{yy} = \Phi(x, y, u, u_x, u_y).$$

- Here Φ is linear in u, u_x, u_y.
- The classification below uses only a, b, c.



Important Examples

- $u_{tt} = k^2 u_{xx}$ one-dimensional wave equation.
- $u_t = k^2 u_{xx}$ one-dimensional heat equation.
- $u_{xx} + u_{yy} = 0$ two-dimensional Laplace equation.
- $u_{xx} + u_{yy} = f(x, y)$ two-dimensional Poisson equation.
- $u_{tt} = k^2(u_{xx} + u_{yy})$ two-dimensional wave equation.
- Wave: vibrating strings, sound and electromagnetic waves.
- Heat: diffusion of heat or of a chemical concentration.
- Laplace / Poisson: steady state temperature, electrostatic and gravitational potentials.



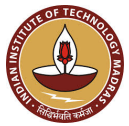
Homogeneous Equations and Superposition

- Equation (1) is homogeneous if $g(x, y) \equiv 0$; otherwise it is non-homogeneous.
- Write $L[u] = a\,u_{xx} + b\,u_{xy} + c\,u_{yy} + d\,u_x + e\,u_y + f\,u$.
- L is a linear operator: $L[c_1 u_1 + c_2 u_2] = c_1 L[u_1] + c_2 L[u_2]$.
- Superposition principle: if u_1, u_2 solve $L[u] = 0$, then so does $c_1 u_1 + c_2 u_2$.
- This is exactly as for linear homogeneous ODEs.
- If u_p solves $L[u] = g$ and u_h solves $L[u] = 0$, then $u_h + u_p$ solves $L[u] = g$.
- Example: $u = x^2 - y^2$ and $u = xy$ solve $u_{xx} + u_{yy} = 0$.
- Hence $u = 3(x^2 - y^2) - 5xy$ also solves the Laplace equation.



ODE Method

- Sometimes ODE methods solve a second order PDE directly.
- This works when derivatives in only one variable appear.
- Or when a substitution such as $v = u_x$ reduces the order.
- Integrating w.r.t. x keeps y fixed.
- So constants of integration become arbitrary functions of y.
- In a second order PDE, two arbitrary functions are expected.
- Such cases are rare, but they motivate the search for simple forms.



Example 1: $u_{xx} - u = 0$

- Only x-derivatives occur; treat y as a parameter.
- For fixed y, this is the ODE $U'' - U = 0$ with roots ± 1.
- Integrating w.r.t. x:

$$u(x, y) = \phi(y)\, e^{x} + \psi(y)\, e^{-x}.$$

- The constants of integration are now functions of y.
- The functions $\phi(y)$ and $\psi(y)$ are arbitrary.
- Check: $u_{xx} = \phi e^{x} + \psi e^{-x} = u$.
- Two arbitrary functions, as expected for a second order PDE.



Example 2: $u_{xy} + u_x = 0$

- Put $v = u_x$. Then the PDE becomes $v_y = -v$.
- Its solution is $v = \phi_1(x)\, e^{-y}$, where $\phi_1(x)$ is the constant of integration.
- Now $u_x = \phi_1(x)\, e^{-y}$ gives

$$u = e^{-y} \int \phi_1(x)\, dx + \psi(y).$$

- ϕ_1 is arbitrary, so its integral is an arbitrary function $\phi(x)$.
- General solution:

$$u(x, y) = e^{-y}\phi(x) + \psi(y).$$

- Check: $u_x = e^{-y}\phi'$, $u_{xy} = -e^{-y}\phi'$, so $u_{xy} + u_x = 0$.



Practice: ODE Method

- Solve $u_{yy} + 4u = 0$. Ans: $u = \phi(x)\cos 2y + \psi(x)\sin 2y$.
- Solve $u_{xx} = 6xy$.
- Solve $u_{xy} = 0$. Ans: $u = \phi(x) + \psi(y)$.
- Solve $u_{xy} - u_y = 0$ (put $v = u_y$).
- Solve $u_{xx} - 3u_x + 2u = 0$.
- Solve $u_{xy} = e^{x+y}$ with $u(x, 0) = 0$, $u(0, y) = 0$.
- In each case identify the two arbitrary functions.



Forms that are Easy to Integrate

- The ODE method suggests transforming (1) into one of the forms

$$u_{xx} = \Phi(x, y, u, u_x, u_y), \qquad u_{xy} = \Phi(x, y, u, u_x, u_y).$$

- Here Φ is linear in u, u_x, u_y:

$$\Phi = f_1(x, y) + f_2(x, y)\, u + f_3(x, y)\, u_x + f_4(x, y)\, u_y.$$

- However, not every linear second order PDE can be brought to these two forms.
- The remaining equations can be transformed to

$$u_{xx} + u_{yy} = \Phi(x, y, u, u_x, u_y).$$

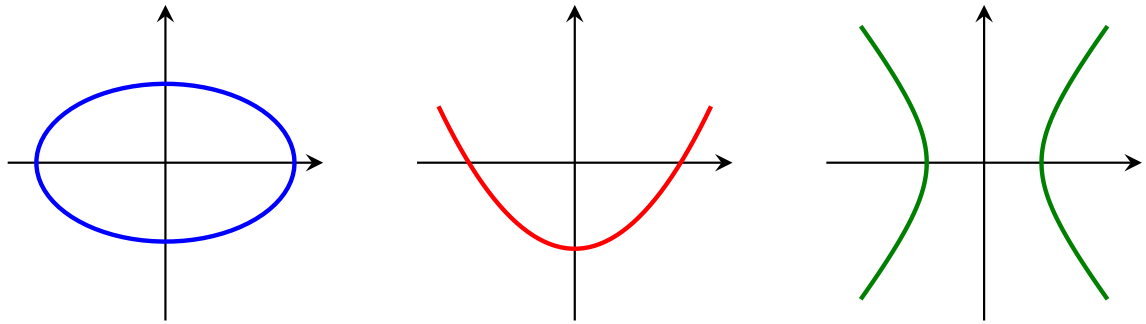
- These are the standard (normal, canonical) forms.
- Which form is possible depends on the type of the PDE.



Analogy with Conic Sections

- Replace u_{xx}, u_{xy}, u_{yy}, u_x, u_y by X^2, XY, Y^2, X, Y (formally).
- The principal part gives the quadratic form $aX^2 + bXY + cY^2$.
- The curve $aX^2 + bXY + cY^2 + dX + eY + f = 0$ is a conic.

ellipse: $b^2 < 4ac$ parabola: $b^2 = 4ac$ hyperbola: $b^2 > 4ac$



- The PDE borrows its name from the type of this conic.

Discriminant and Types

- The discriminant of (1) is $b^2 - 4ac$ (it involves only the principal part).
- At a point (x_0, y_0), the PDE (1) is called.
- hyperbolic if $b^2 - 4ac > 0$,.
- parabolic if $b^2 - 4ac = 0$,.
- elliptic if $b^2 - 4ac < 0$.
- The PDE is of a given type in a region if it is of that type at every point there.
- Caution: $b^2 - 4ac$ is a function of x and y.
- Its sign may change, so one PDE can be of different types in different regions.



Standard Forms and Their Types

- Look at the signs of the discriminants of the standard forms (Φ = lower order terms).

Canonical form	$a,\ b,\ c$	$b^2 - 4ac$	Type
$u_{xy} = \Phi$	$a = 0,\ b = 1,\ c = 0$	> 0	Hyperbolic
$u_{xx} - u_{yy} = \Phi$	$a = 1,\ b = 0,\ c = -1$	> 0	Hyperbolic
$u_{xx} = \Phi$	$a = 1,\ b = 0,\ c = 0$	$= 0$	Parabolic
$u_{xx} + u_{yy} = \Phi$	$a = 1,\ b = 0,\ c = 1$	< 0	Elliptic

- Key fact: the sign of the discriminant does not change under a smooth change of independent variables.
- Hence only a hyperbolic equation can be reduced to $u_{\xi\eta} = \Phi$, and so on.
- We will prove this fact shortly.



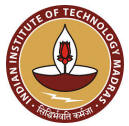
Classifying the Basic Equations

- Wave equation $u_{tt} - k^2 u_{xx} = 0$ in (x, t): $a = -k^2$, $b = 0$, $c = 1$.
- $b^2 - 4ac = 4k^2 > 0$: hyperbolic everywhere.
- Heat equation $k^2 u_{xx} - u_t = 0$: $a = k^2$, $b = 0$, $c = 0$.
- $b^2 - 4ac = 0$: parabolic everywhere.
- Laplace equation $u_{xx} + u_{yy} = 0$: $a = 1$, $b = 0$, $c = 1$.
- $b^2 - 4ac = -4 < 0$: elliptic everywhere.
- Hyperbolic: propagation of waves; parabolic: diffusion; elliptic: equilibrium.
- Lower order terms (u_t in the heat equation) do not affect the type.

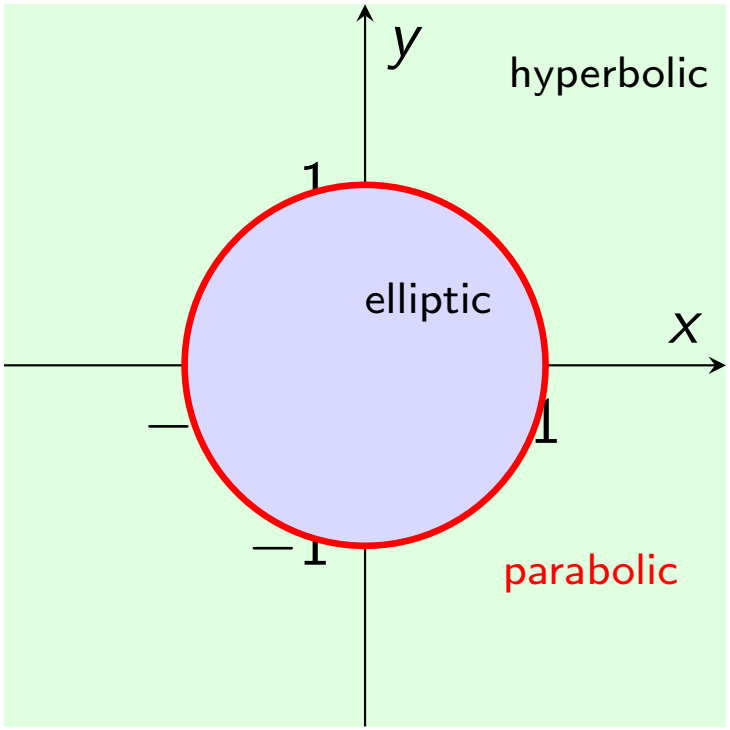


Example: Type Depending on the Region

- Find where $u_{xx} + 2x\,u_{xy} + (1 - y^2)\,u_{yy} = 0$ is hyperbolic, parabolic or elliptic.
- Here $a = 1$, $b = 2x$, $c = 1 - y^2$.
- $b^2 - 4ac = 4x^2 - 4(1 - y^2) = 4(x^2 + y^2 - 1)$.
- Hyperbolic if $x^2 + y^2 > 1$ (outside the unit circle).
- Parabolic if $x^2 + y^2 = 1$ (on the unit circle).
- Elliptic if $x^2 + y^2 < 1$ (inside the unit circle).
- The boundary between the elliptic and hyperbolic regions is the parabolic curve.



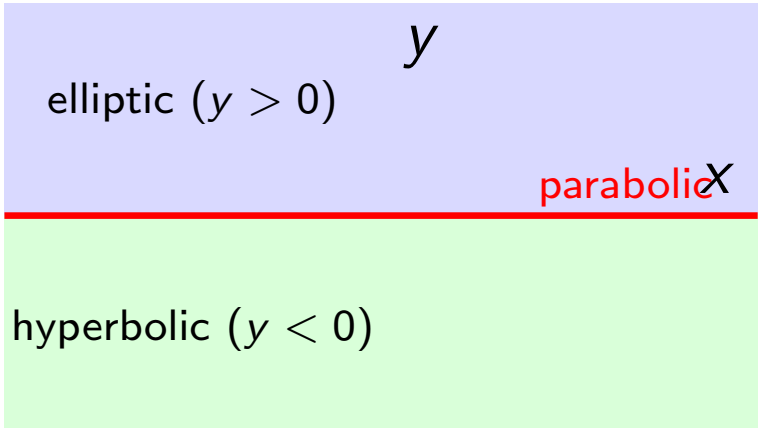
Picture: Regions of Different Types



- $u_{xx} + 2x\,u_{xy} + (1 - y^2)\,u_{yy} = 0$: elliptic inside, parabolic on, hyperbolic outside the unit circle.

Example: The Tricomi Equation

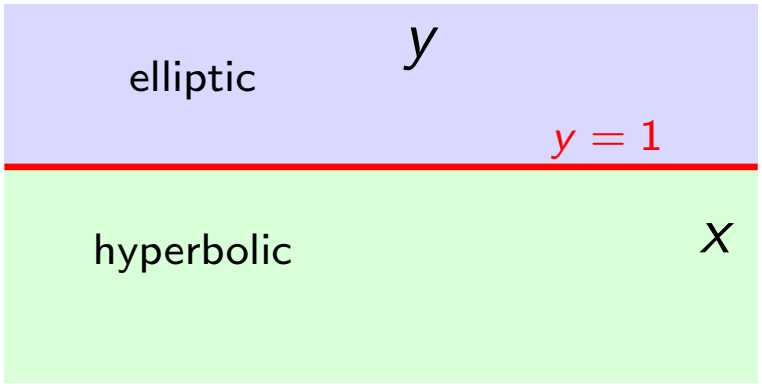
- $y\,u_{xx} + u_{yy} = 0$ arises in transonic gas dynamics.
- $a = y$, $b = 0$, $c = 1$, so $b^2 - 4ac = -4y$.



- Elliptic for $y > 0$, parabolic on $y = 0$, hyperbolic for $y < 0$.

Example: $y\,u_{xx} + 2u_{xy} + u_{yy} = u\,u_x + u_y^2$

- The right side is nonlinear, but the principal part is linear; classify by a, b, c.
- $a = y,\ b = 2,\ c = 1$: $\quad b^2 - 4ac = 4 - 4y = 4(1 - y)$.



- Hyperbolic for $y < 1$, parabolic on $y = 1$, elliptic for $y > 1$.

Practice: Classification

- Classify the following PDEs and sketch the regions:
- (1) $u_{xx} + 4u_{xy} + 4u_{yy} = 0.$ (parabolic).
- (2) $x^2 u_{xx} - y^2 u_{yy} = 0.$ (hyperbolic if $xy \neq 0$).
- (3) $u_{xx} + x\, u_{yy} = 0.$
- (4) $x\, u_{xx} + y\, u_{yy} + 3y^2 u_x = 0.$
- (5) $(1 + x^2)u_{xx} + (1 + y^2)u_{yy} + x\, u_x + y\, u_y = 0.$
- (6) $\sin^2 x\, u_{xx} + \sin 2x\, u_{xy} + \cos^2 x\, u_{yy} = x.$
- (7) $u_{xx} + y\, u_{xy} + x\, u_{yy} = 0.$



Reduction to Standard Forms: New Variables

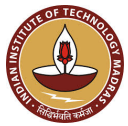
- Introduce new independent variables

$$\xi = \xi(x, y), \qquad \eta = \eta(x, y).$$

- Assume ξ, η have continuous second order partial derivatives.
- Assume the Jacobian is non-zero in the region:

$$J = \frac{\partial(\xi, \eta)}{\partial(x, y)} = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \xi_x \eta_y - \xi_y \eta_x \neq 0.$$

- This guarantees that x, y can be recovered from ξ, η (locally).
- Then $u(x, y) = u\big(x(\xi, \eta),\, y(\xi, \eta)\big)$ is regarded as a function of ξ, η.
- Aim: choose ξ, η so that the new equation is as simple as possible.



First Derivatives

- By the chain rule,

$$u_x = u_\xi \, \xi_x + u_\eta \, \eta_x, \tag{2}$$

$$u_y = u_\xi \, \xi_y + u_\eta \, \eta_y. \tag{3}$$

- Note that u_ξ, u_η are again functions of ξ, η.
- So differentiating them w.r.t. x again needs the chain rule:

$$\frac{\partial}{\partial x}(u_\xi) = u_{\xi\xi} \, \xi_x + u_{\xi\eta} \, \eta_x, \qquad \frac{\partial}{\partial x}(u_\eta) = u_{\eta\xi} \, \xi_x + u_{\eta\eta} \, \eta_x.$$

- Also use the product rule on $u_\xi \, \xi_x$ and $u_\eta \, \eta_x$.



Second Derivatives: u_{xx} and u_{yy}

- From (2),

$$u_{xx} = (u_{\xi\xi}\xi_x + u_{\xi\eta}\eta_x)\xi_x + u_\xi\,\xi_{xx} + (u_{\eta\xi}\xi_x + u_{\eta\eta}\eta_x)\eta_x + u_\eta\,\eta_{xx}.$$

- Hence

$$u_{xx} = u_{\xi\xi}\,\xi_x^2 + 2u_{\xi\eta}\,\xi_x\eta_x + u_{\eta\eta}\,\eta_x^2 + u_\xi\,\xi_{xx} + u_\eta\,\eta_{xx}. \qquad (4)$$

- Similarly, from (3),

$$u_{yy} = u_{\xi\xi}\,\xi_y^2 + 2u_{\xi\eta}\,\xi_y\eta_y + u_{\eta\eta}\,\eta_y^2 + u_\xi\,\xi_{yy} + u_\eta\,\eta_{yy}. \qquad (5)$$



Second Derivatives: u_{xy}

- Differentiate (2) w.r.t. y:

$$u_{xy} = (u_{\xi\xi}\xi_y + u_{\xi\eta}\eta_y)\xi_x + u_\xi\,\xi_{xy} + (u_{\eta\xi}\xi_y + u_{\eta\eta}\eta_y)\eta_x + u_\eta\,\eta_{xy}.$$

- Hence

$$u_{xy} = u_{\xi\xi}\,\xi_x\xi_y + u_{\xi\eta}(\xi_x\eta_y + \xi_y\eta_x) + u_{\eta\eta}\,\eta_x\eta_y + u_\xi\,\xi_{xy} + u_\eta\,\eta_{xy}. \quad (6)$$

- Formulas (4)–(6) are used in every reduction.
- In examples, it is often quicker to differentiate directly than to substitute into them.



The Transformed Equation

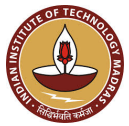
- Substituting (2)–(6) in (1) and grouping terms, we obtain

$$A\,u_{\xi\xi} + B\,u_{\xi\eta} + C\,u_{\eta\eta} + D\,u_{\xi} + E\,u_{\eta} + F\,u = G, \qquad (7)$$

where

$$\begin{aligned}
A &= a\,\xi_x^2 + b\,\xi_x\xi_y + c\,\xi_y^2,\\
B &= 2a\,\xi_x\eta_x + b\,(\xi_x\eta_y + \xi_y\eta_x) + 2c\,\xi_y\eta_y,\\
C &= a\,\eta_x^2 + b\,\eta_x\eta_y + c\,\eta_y^2,\\
D &= a\,\xi_{xx} + b\,\xi_{xy} + c\,\xi_{yy} + d\,\xi_x + e\,\xi_y,\\
E &= a\,\eta_{xx} + b\,\eta_{xy} + c\,\eta_{yy} + d\,\eta_x + e\,\eta_y,\\
F &= f, \qquad G = g.
\end{aligned}$$

- All coefficients are finally expressed in terms of ξ and η.



Observations on the New Coefficients

- $a, \ldots, e$ must first be expressed in terms of ξ, η; then so are $A, \ldots, E$.
- F and G do not change; they are only rewritten in terms of ξ, η.
- A, B, C depend only on a, b, c and first derivatives of ξ, η.
- So the new principal part depends only on the old principal part.
- A and C have the same form: $A = Q(\xi_x, \xi_y)$, $C = Q(\eta_x, \eta_y)$, where
$$Q(\lambda, \mu) = a\lambda^2 + b\lambda\mu + c\mu^2.$$
- The equation (7) is again second order linear, with the same structure.
- Next: compare the discriminants $b^2 - 4ac$ and $B^2 - 4AC$.



Invariance of the Discriminant

- Theorem. The discriminants of (1) and (7) are related by

$$B^2 - 4AC = J^2\,(b^2 - 4ac), \qquad J = \xi_x \eta_y - \xi_y \eta_x.$$

- Since $J \neq 0$, we have $J^2 > 0$.
- Hence $B^2 - 4AC$ and $b^2 - 4ac$ have the same sign.
- The type of the PDE remains the same under such a general transformation.
- So the classification is meaningful: it is a property of the equation, not of the coordinates.
- Exercise: verify the identity by direct expansion.



Proof using Matrices

- Write the principal parts with symmetric matrices:

$$M = \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}, \qquad N = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}.$$

- Let $\mathcal{J} = \begin{pmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{pmatrix}$, so that $\det \mathcal{J} = J$.
- The formulas for A, B, C say exactly that $N = \mathcal{J}\, M\, \mathcal{J}^T$.
- Taking determinants: $\det N = (\det \mathcal{J})^2 \det M = J^2 \det M$.
- But $\det N = AC - \frac{B^2}{4}$ and $\det M = ac - \frac{b^2}{4}$.
- Multiplying by -4: $B^2 - 4AC = J^2(b^2 - 4ac)$.



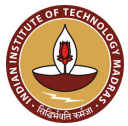
Factorization of A and C

- We must choose ξ and η depending on the type.
- If $a \neq 0$, the quadratic forms A and C factor as

$$A = \frac{1}{4a}\left[2a\xi_x + \left(b + \sqrt{b^2 - 4ac}\right)\xi_y\right]\left[2a\xi_x + \left(b - \sqrt{b^2 - 4ac}\right)\xi_y\right],$$

$$C = \frac{1}{4a}\left[2a\eta_x + \left(b + \sqrt{b^2 - 4ac}\right)\eta_y\right]\left[2a\eta_x + \left(b - \sqrt{b^2 - 4ac}\right)\eta_y\right].$$

- Check: the product of the brackets is $4a^2\xi_x^2 + 4ab\,\xi_x\xi_y + 4ac\,\xi_y^2 = 4a\,A$.
- To make $A = 0$, it suffices to make one factor zero: a first order linear PDE for ξ.
- The factors are real, equal or complex according as the PDE is hyperbolic, parabolic or elliptic.



Characteristic Equation

- We look for ξ with $A = a\,\xi_x^2 + b\,\xi_x\xi_y + c\,\xi_y^2 = 0$, i.e.

$$a\left(\frac{\xi_x}{\xi_y}\right)^2 + b\left(\frac{\xi_x}{\xi_y}\right) + c = 0. \tag{8}$$

- Consider a level curve $\xi(x, y) = c_1$. Differentiating w.r.t. x:

$$\xi_x + \xi_y \frac{dy}{dx} = 0 \;\Rightarrow\; \frac{\xi_x}{\xi_y} = -\frac{dy}{dx}. \tag{9}$$

- Substituting (9) in (8):

$$a\left(\frac{dy}{dx}\right)^2 - b\,\frac{dy}{dx} + c = 0.$$

- This is the characteristic equation of (1).



Characteristic Curves

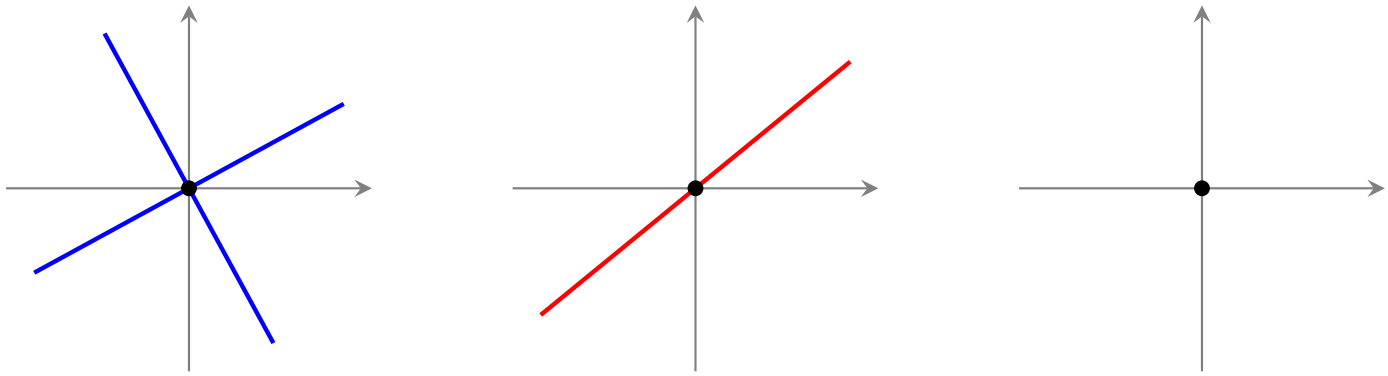
- Solving the characteristic equation (for $a \neq 0$):

$$\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{10}$$

- The integral curves of (10) are the characteristic curves (characteristics) of (1).
- If $\xi(x, y) = c_1$ is an integral curve of (10), then ξ makes $A = 0$.
- Similarly, if $\eta(x, y) = c_2$ is an integral curve, then η makes $C = 0$.
- Equivalent symmetric form: $a\,(dy)^2 - b\,dx\,dy + c\,(dx)^2 = 0$.
- If $a = 0$ but $c \neq 0$, solve for dx/dy instead.
- Hyperbolic: two real families; parabolic: one real family; elliptic: no real family.



Picture: Characteristic Directions at a Point



hyperbolic: two real parabolic: one (double) elliptic: none real

- At each point, (10) gives the slopes of the characteristic directions.
- Their number (2, 1, 0) is decided by the sign of $b^2 - 4ac$.



Geometric Meaning of Characteristics

- Let u, u_x, u_y be prescribed along a curve $\Gamma : x = x(s)$, $y = y(s)$ (Cauchy data).
- Can we find u_{xx}, u_{xy}, u_{yy} on Γ? Along Γ:

$$d(u_x) = u_{xx}\,dx + u_{xy}\,dy, \qquad d(u_y) = u_{xy}\,dx + u_{yy}\,dy.$$

- Together with the PDE, $a\,u_{xx} + b\,u_{xy} + c\,u_{yy} = \Phi$ (known on Γ).
- A linear system of three equations for u_{xx}, u_{xy}, u_{yy} with determinant

$$\begin{vmatrix} a & b & c \\ dx & dy & 0 \\ 0 & dx & dy \end{vmatrix} = a\,(dy)^2 - b\,dx\,dy + c\,(dx)^2.$$

- It is uniquely solvable unless $a\,(dy)^2 - b\,dx\,dy + c\,(dx)^2 = 0$.

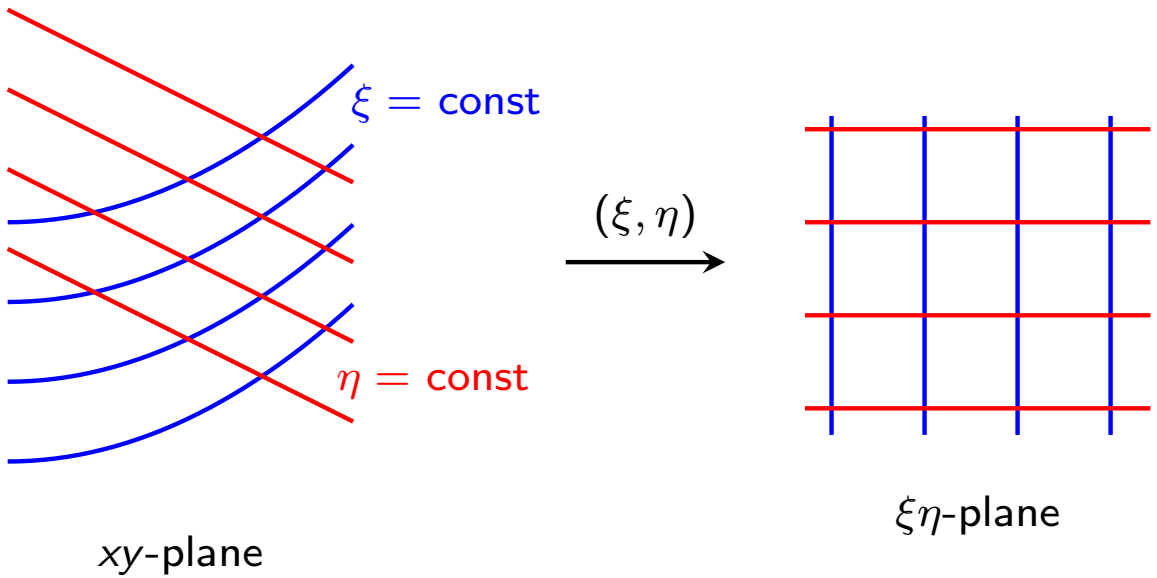


Geometric Meaning (contd.)

- The exceptional directions are exactly the characteristic directions.
- On a non-characteristic curve, Cauchy data determine all second (and higher) derivatives.
- On a characteristic curve, the second derivatives are not determined by the data.
- Hence second derivatives of a solution may jump across characteristics.
- Discontinuities (e.g. wave fronts) propagate along characteristics.
- This is the same role characteristics played for first order PDEs.
- Elliptic equations have no real characteristics: their solutions are smooth.
- Hyperbolic equations admit wave fronts travelling along the two families.



Picture: Characteristic Coordinates



- Characteristic curves become coordinate lines in the new variables.
- In these variables the equation takes its simplest (canonical) form.



Hyperbolic Type: Choice of Variables

- Suppose $b^2 - 4ac > 0$ in the region of interest.
- If $a = 0 = c$, the PDE is already $b\,u_{xy} = \Phi$, a standard form.
- Otherwise assume $a \neq 0$ (else interchange the roles of x and y).
- We try to make both $A = 0$ and $C = 0$, with $B \neq 0$.
- Take different factors in the factorizations of A and C:

$$2a\,\xi_x + \left(b + \sqrt{b^2 - 4ac}\right)\xi_y = 0, \qquad 2a\,\eta_x + \left(b - \sqrt{b^2 - 4ac}\right)\eta_y = 0.$$

- By Lagrange's method these first order PDEs have characteristic equations

$$\frac{dx}{2a} = \frac{dy}{b + \sqrt{b^2 - 4ac}}, \qquad \frac{dx}{2a} = \frac{dy}{b - \sqrt{b^2 - 4ac}}. \tag{11}$$



Hyperbolic Type: Two Families

- Equations (11) are the two real equations in (10):

$$\frac{dy}{dx} = \frac{b + \sqrt{b^2 - 4ac}}{2a}, \qquad \frac{dy}{dx} = \frac{b - \sqrt{b^2 - 4ac}}{2a}.$$

- Let their solutions be $\phi(x, y) = c_1$ and $\psi(x, y) = c_2$.
- Take the transformation $\xi = \phi(x, y)$, $\eta = \psi(x, y)$.
- Then $A = 0$ and $C = 0$ by construction.
- The slopes differ ($\sqrt{b^2 - 4ac} \neq 0$), so the two families are transversal and $J \neq 0$.
- From $B^2 - 4AC = J^2(b^2 - 4ac)$ we get $B^2 = J^2(b^2 - 4ac) > 0$, so $B \neq 0$.



First Canonical Form of a Hyperbolic PDE

- With $A = C = 0$, equation (7) becomes

$$B\, u_{\xi\eta} + D\, u_{\xi} + E\, u_{\eta} + F\, u = G.$$

- Dividing by $B \neq 0$:

$$u_{\xi\eta} = \frac{1}{B}\left(G - D\, u_{\xi} - E\, u_{\eta} - F\, u\right) = \Phi(\xi, \eta, u, u_{\xi}, u_{\eta}).$$

- This is called the first canonical form of a hyperbolic PDE.
- The variables ξ, η are called characteristic coordinates.
- If $\Phi = 0$, the general solution is $u = \phi(\xi) + \psi(\eta)$.
- Note: the choice of ξ, η is not unique; any $\tilde{\xi} = h_1(\xi)$, $\tilde{\eta} = h_2(\eta)$ also works.



Second Canonical Form of a Hyperbolic PDE

- Starting from $u_{\xi\eta} = \Phi$, introduce

$$\alpha = \xi + \eta, \qquad \beta = \xi - \eta.$$

- Then $u_\xi = u_\alpha + u_\beta, \;\; u_\eta = u_\alpha - u_\beta$.
- Differentiating again: $u_{\xi\eta} = u_{\alpha\alpha} - u_{\beta\beta}$.
- Hence

$$u_{\alpha\alpha} - u_{\beta\beta} = \Phi_1(\alpha, \beta, u, u_\alpha, u_\beta).$$

- This is the second canonical form of a hyperbolic PDE.
- It resembles the wave equation $u_{xx} - u_{yy} = 0$.
- Both forms are useful: the first for integration, the second for comparison with physics.



Working Procedure (Hyperbolic)

- Step 1: Identify a, b, c and check $b^2 - 4ac > 0$ in the region.
- Step 2: Write $\dfrac{dy}{dx} = \dfrac{b \pm \sqrt{b^2 - 4ac}}{2a}$.
- Step 3: Integrate to get $\phi(x, y) = c_1$ and $\psi(x, y) = c_2$.
- Step 4: Put $\xi = \phi(x, y)$, $\eta = \psi(x, y)$; check $J \neq 0$.
- Step 5: Compute u_x, u_y, u_{xx}, u_{xy}, u_{yy} in terms of ξ, η.
- Step 6: Substitute; the $u_{\xi\xi}$ and $u_{\eta\eta}$ terms must cancel.
- Step 7: Express the remaining coefficients in ξ, η and divide by B.
- Step 8: If possible, integrate the canonical form to get the general solution.



Example: $3u_{xx} + 10u_{xy} + 3u_{yy} = 0$

- $a = 3,\ b = 10,\ c = 3$: $b^2 - 4ac = 100 - 36 = 64 > 0$, hyperbolic everywhere.
- Characteristic equation: $\dfrac{dy}{dx} = \dfrac{10 \pm 8}{6} = 3$ or $\dfrac{1}{3}$.
- $dy = 3\,dx \Rightarrow y - 3x = c_1$; $\quad 3\,dy = dx \Rightarrow 3y - x = c_2$.
- Put $\xi = y - 3x,\ \eta = 3y - x$; $\ J = (-3)(3) - (1)(-1) = -8 \neq 0$.
- $u_x = -3u_\xi - u_\eta,\ \ u_y = u_\xi + 3u_\eta$.
- $u_{xx} = 9u_{\xi\xi} + 6u_{\xi\eta} + u_{\eta\eta},\ \ u_{xy} = -3u_{\xi\xi} - 10u_{\xi\eta} - 3u_{\eta\eta}$.
- $u_{yy} = u_{\xi\xi} + 6u_{\xi\eta} + 9u_{\eta\eta}$.



Example (contd.)

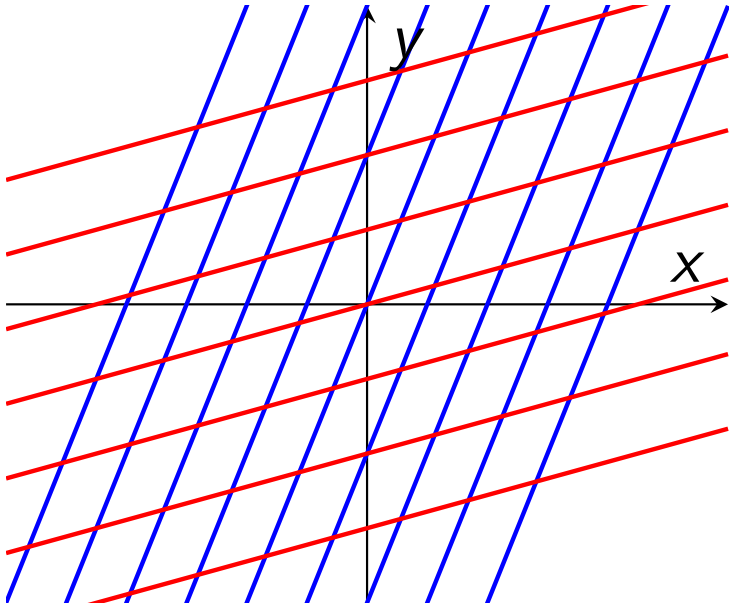
- Substituting:

$$3(9u_{\xi\xi}+6u_{\xi\eta}+u_{\eta\eta})+10(-3u_{\xi\xi}-10u_{\xi\eta}-3u_{\eta\eta})+3(u_{\xi\xi}+6u_{\xi\eta}+9u_{\eta\eta}) =$$

- Coefficients of $u_{\xi\xi}$ and $u_{\eta\eta}$: $27 - 30 + 3 = 0$.
- Coefficient of $u_{\xi\eta}$: $18 - 100 + 18 = -64$.
- Canonical form: $-64\, u_{\xi\eta} = 0$, i.e. $u_{\xi\eta} = 0$.
- Check: $B^2 - 4AC = 64^2 = J^2(b^2 - 4ac) = 64 \times 64$.
- General solution: $u(x, y) = \phi(y - 3x) + \psi(3y - x)$, with ϕ, ψ arbitrary.



Picture: Characteristics of $3u_{xx} + 10u_{xy} + 3u_{yy} = 0$



- Two families of straight lines: $y - 3x = c_1$ (blue) and $3y - x = c_2$ (red).



Example: $u_{xx} - y^4 u_{yy} = 2y^3 u_y, \quad y \neq 0$

- $a = 1,\ b = 0,\ c = -y^4$: $\quad b^2 - 4ac = 4y^4 > 0$ for $y \neq 0$; hyperbolic.
- Characteristic equations: $\dfrac{dy}{dx} = \dfrac{0 \pm 2y^2}{2} = \pm y^2.$
- $\dfrac{dy}{y^2} = dx \Rightarrow -\dfrac{1}{y} = x + c_1; \quad -\dfrac{dy}{y^2} = dx \Rightarrow \dfrac{1}{y} = x + c_2.$
- Put $\xi = x + \dfrac{1}{y}, \quad \eta = x - \dfrac{1}{y}.$
- $\xi_x = 1,\ \xi_y = -\dfrac{1}{y^2},\ \eta_x = 1,\ \eta_y = \dfrac{1}{y^2}; \quad J = \dfrac{2}{y^2} \neq 0.$
- $u_x = u_\xi + u_\eta, \quad u_{xx} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}.$



Example (contd.)

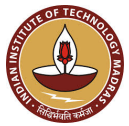
- $u_y = u_\xi\,\xi_y + u_\eta\,\eta_y = \dfrac{1}{y^2}\big(-u_\xi + u_\eta\big)$.
- Differentiating again w.r.t. y (product and chain rules):

$$u_{yy} = \frac{1}{y^4}\big(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}\big) + \frac{2}{y^3}\big(u_\xi - u_\eta\big).$$

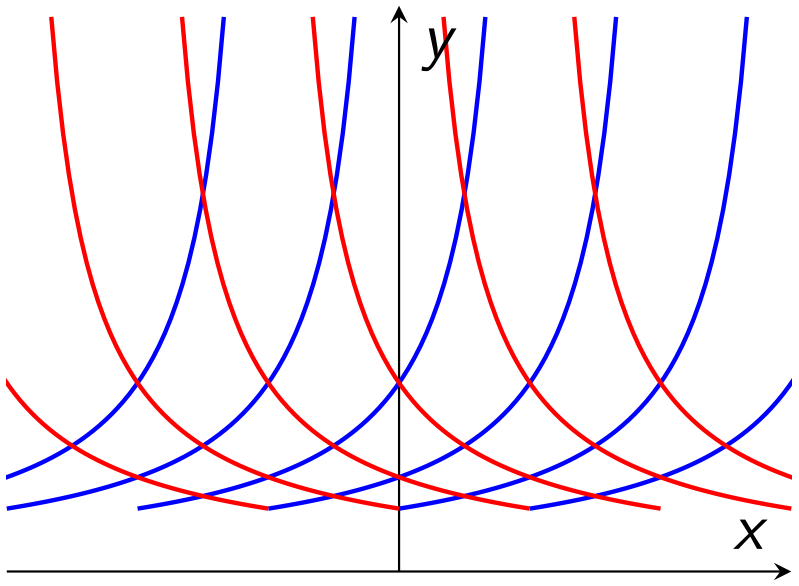
- Substitute in $u_{xx} - y^4 u_{yy} = 2y^3 u_y$:

$$(u_{\xi\xi}+2u_{\xi\eta}+u_{\eta\eta})-(u_{\xi\xi}-2u_{\xi\eta}+u_{\eta\eta})-2y(u_\xi-u_\eta) = 2y(-u_\xi+u_\eta).$$

- The first order terms cancel: $4u_{\xi\eta} = 0$.
- Canonical form: $u_{\xi\eta} = 0$.
- General solution: $u(x,y) = \phi\big(x + \frac{1}{y}\big) + \psi\big(x - \frac{1}{y}\big)$.



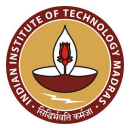
Picture: Characteristics $x \pm 1/y = \text{const}$



- In $y > 0$: $y = \dfrac{1}{c_1 - x}$ (blue, $\xi = c_1$) and $y = \dfrac{1}{x - c_2}$ (red, $\eta = c_2$).

Example: $u_{xx} - x\,u_{yy} = \dfrac{u_x}{2x},\quad x > 0$

- $a = 1,\ b = 0,\ c = -x$: $\quad b^2 - 4ac = 4x > 0$, hyperbolic for $x > 0$.
- Characteristic equations: $\dfrac{dy}{dx} = \pm\sqrt{x}$.
- $dy = \pm\sqrt{x}\,dx \Rightarrow y \mp \frac{2}{3}x^{3/2} = \text{const.}$
- Put $\xi = y - \frac{2}{3}x^{3/2},\ \ \eta = y + \frac{2}{3}x^{3/2}$.
- $\xi_x = -\sqrt{x},\ \xi_y = 1,\ \eta_x = \sqrt{x},\ \eta_y = 1;\ \ J = -2\sqrt{x} \neq 0$.
- $u_x = \sqrt{x}\,(-u_\xi + u_\eta)$.
- $u_y = u_\xi + u_\eta,\ \ u_{yy} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$.



Example (contd.)

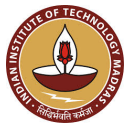
- Differentiate $u_x = \sqrt{x}\,(-u_\xi + u_\eta)$ w.r.t. x:

$$u_{xx} = x\,(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}) + \frac{1}{2\sqrt{x}}\,(-u_\xi + u_\eta).$$

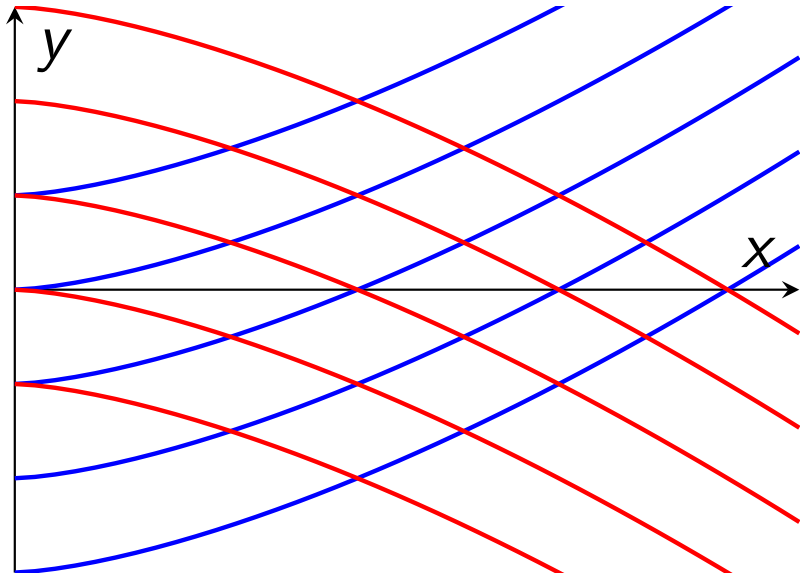
- Substitute in the PDE:

$$x(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}) + \frac{-u_\xi + u_\eta}{2\sqrt{x}} - x(u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}) = \frac{\sqrt{x}\,(-u_\xi + u_\eta)}{2x}$$

- The first order terms cancel, leaving $-4x\,u_{\xi\eta} = 0$.
- Canonical form: $u_{\xi\eta} = 0$.
- General solution: $u(x, y) = \phi\left(y - \frac{2}{3}x^{3/2}\right) + \psi\left(y + \frac{2}{3}x^{3/2}\right)$.



Picture: Characteristics $y \mp \frac{2}{3}x^{3/2} = \text{const}$



- Two families of semicubical parabolas; both become horizontal on $x = 0$, where the PDE is parabolic.

Example: The Wave Equation $u_{tt} = k^2 u_{xx}$

- Write it as $u_{tt} - k^2 u_{xx} = 0$ in the variables (t, x) (so t plays the role of x).
- $a = 1,\ b = 0,\ c = -k^2$: $\quad b^2 - 4ac = 4k^2 > 0$, hyperbolic.
- Characteristic equations: $\dfrac{dx}{dt} = \pm k \Rightarrow x \mp kt = \text{const.}$
- Put $\xi = x + kt,\ \ \eta = x - kt;\ \ J = \xi_t \eta_x - \xi_x \eta_t = k + k = 2k \neq 0.$
- $u_t = k\, u_\xi - k\, u_\eta,\ \ u_{tt} = k^2(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}).$
- $u_x = u_\xi + u_\eta,\ \ u_{xx} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}.$
- Substituting:
  $k^2(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}) - k^2(u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}) = -4k^2 u_{\xi\eta} = 0.$

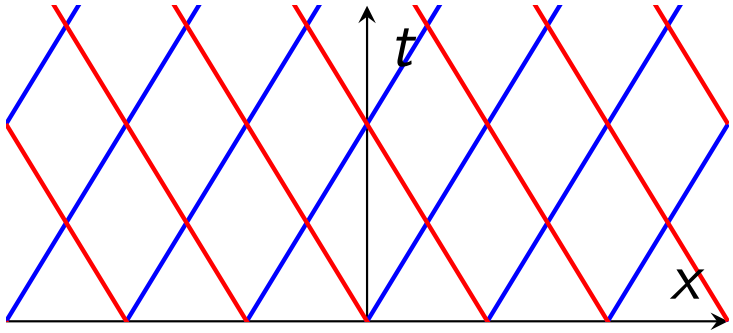


Wave Equation: Canonical Form and General Solution

- Canonical form: $u_{\xi\eta} = 0$.
- Integrating: $u(\xi, \eta) = \phi(\xi) + \psi(\eta)$.
- General solution:

$$u(x, t) = \phi(x + kt) + \psi(x - kt),$$

where ϕ and ψ are arbitrary twice differentiable functions.



- Characteristics: straight lines $x - kt = c_2$ (blue) and $x + kt = c_1$ (red); here $k = 1$.

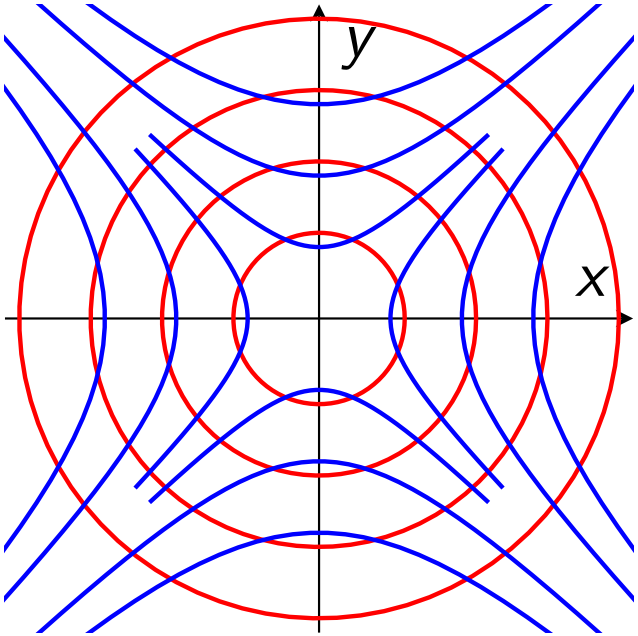
Example: $y^2 u_{xx} - x^2 u_{yy} = 0, \;\; x, y \neq 0$

- $a = y^2$, $b = 0$, $c = -x^2$: $\;\; b^2 - 4ac = 4x^2y^2 > 0$, hyperbolic for $xy \neq 0$.
- Characteristic equations: $\dfrac{dy}{dx} = \pm\dfrac{x}{y} \Rightarrow y\,dy \mp x\,dx = 0$.
- Integrating: $y^2 - x^2 = c_1$ and $y^2 + x^2 = c_2$.
- Put $\xi = y^2 - x^2$, $\;\; \eta = y^2 + x^2$;
  $J = (-2x)(2y) - (2y)(2x) = -8xy \neq 0$.
- $A = C = 0$, $\;\; B = -16x^2y^2$, $\;\; D = -2y^2 - 2x^2$, $\;\; E = 2y^2 - 2x^2$.
- Using $x^2 = \frac{\eta - \xi}{2}$, $y^2 = \frac{\eta + \xi}{2}$: $\;\; B = -4(\eta^2 - \xi^2)$, $D = -2\eta$, $E = 2\xi$.
- Canonical form:

$$u_{\xi\eta} = \frac{\eta\, u_\xi - \xi\, u_\eta}{2(\xi^2 - \eta^2)}.$$



Picture: Characteristics of $y^2 u_{xx} - x^2 u_{yy} = 0$



- Circles $x^2 + y^2 = c_2$ (red) and rectangular hyperbolas $y^2 - x^2 = c_1$ (blue).



Example: Tricomi Equation for $y < 0$

- $y\,u_{xx} + u_{yy} = 0$ is hyperbolic for $y < 0$ (here $a = y$, $b = 0$, $c = 1$).
- Characteristic equations: $\dfrac{dy}{dx} = \dfrac{\pm\sqrt{-4y}}{2y} = \mp\dfrac{1}{\sqrt{-y}}$.
- $\sqrt{-y}\,dy = \mp dx \Rightarrow x \mp \frac{2}{3}(-y)^{3/2} = \text{const.}$
- Put $\xi = x - \frac{2}{3}(-y)^{3/2}$, $\eta = x + \frac{2}{3}(-y)^{3/2}$.
- A computation (as before) gives $A = C = 0$, $B = 4y$, $D = -E = -\dfrac{1}{2\sqrt{-y}}$.
- With $\xi - \eta = -\frac{4}{3}(-y)^{3/2}$, the canonical form is

$$u_{\xi\eta} = \frac{u_\xi - u_\eta}{6(\xi - \eta)}.$$

- The elliptic part $y > 0$ is treated later.



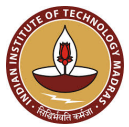
Parabolic Type: Choice of ξ

- Suppose $b^2 - 4ac = 0$, i.e. $b^2 = 4ac$, in the region; assume $a \neq 0$.
- Only one real family of characteristics exists, so we can make only $A = 0$ (or $C = 0$).
- For $A = 0$: $a\,\xi_x^2 + b\,\xi_x\xi_y + c\,\xi_y^2 = 0$.
- Multiply by $4a$ and use $4ac = b^2$:

$$4a^2\xi_x^2 + 4ab\,\xi_x\xi_y + b^2\xi_y^2 = 0 \;\Rightarrow\; (2a\,\xi_x + b\,\xi_y)^2 = 0. \qquad (12)$$

- With $\xi(x, y) = c_1$, $\xi_x/\xi_y = -dy/dx$, so $\left(-2a\,\dfrac{dy}{dx} + b\right)^2 = 0$.
- Characteristic equation (double root of (10)):

$$\frac{dy}{dx} = \frac{b}{2a}.$$



Parabolic Type: B Vanishes Too

- Let $\xi(x,y) = c_1$ solve $dy/dx = b/2a$. Then $2a\xi_x + b\xi_y = 0$ by (12).
- Choose any $\eta(x,y)$ with $J = \xi_x\eta_y - \xi_y\eta_x \neq 0$ (often $\eta = x$ or y).
- Then

$$B = (2a\,\xi_x + b\,\xi_y)\,\eta_x + (b\,\xi_x + 2c\,\xi_y)\,\eta_y = (b\,\xi_x + 2c\,\xi_y)\,\eta_y.$$

- Claim: $b\,\xi_x + 2c\,\xi_y = 0$.
- If $b \neq 0$ (so $c \neq 0$ as $b^2 = 4ac$):

$$b\,\xi_x + 2c\,\xi_y = \tfrac{1}{b}\left(b^2\xi_x + 2bc\,\xi_y\right) = \tfrac{1}{b}\left(4ac\,\xi_x + 2bc\,\xi_y\right) = \tfrac{2c}{b}(2a\,\xi_x + b\,\xi$$

- If $b = 0$, then $c = 0$ and the claim is trivial. Hence $B = 0$.
- This also follows from $B^2 - 4AC = 0$ with $A = 0$.



Canonical Form of a Parabolic PDE

- With $A = B = 0$, equation (7) becomes $C\, u_{\eta\eta} + D\, u_{\xi} + E\, u_{\eta} + F\, u = G$.
- $C \neq 0$, otherwise the equation would not be of second order.
- Dividing by C:

$$u_{\eta\eta} = \Phi(\xi, \eta, u, u_{\xi}, u_{\eta}).$$

- This is the canonical form of a parabolic PDE.
- Note: if instead $c \neq 0$ and we take $\eta(x, y) = c_2$ as the characteristic, we get $C = B = 0$ and the form $u_{\xi\xi} = \Phi$.
- The heat equation $u_t = k^2 u_{xx}$ is already in canonical form.
- Only one family of characteristics: $\xi = \text{const}$.



Example: $u_{xx} + 2u_{xy} + u_{yy} = 4xy$

- $a = 1,\ b = 2,\ c = 1$: $b^2 - 4ac = 0$ everywhere, so parabolic in $\mathbb{R}^2$.
- Characteristic equation: $\dfrac{dy}{dx} = \dfrac{b}{2a} = 1 \Rightarrow y - x = c_1$.
- Let $\xi = y - x$ and choose $\eta = y + x$;
  $J = (-1)(1) - (1)(1) = -2 \neq 0$.
- Inverse: $x = \dfrac{\eta - \xi}{2},\ y = \dfrac{\eta + \xi}{2}$.
- $u_x = -u_\xi + u_\eta,\ \ u_{xx} = u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}$.
- $u_{xy} = -u_{\xi\xi} + u_{\eta\eta}$.
- $u_y = u_\xi + u_\eta,\ \ u_{yy} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$.



Example (contd.)

- Substituting in the PDE:

$$(u_{\xi\xi}-2u_{\xi\eta}+u_{\eta\eta})+2(-u_{\xi\xi}+u_{\eta\eta})+(u_{\xi\xi}+2u_{\xi\eta}+u_{\eta\eta}) = 4\cdot\frac{\eta-\xi}{2}\cdot\frac{\eta+\xi}{2}.$$

- $\Rightarrow 4u_{\eta\eta} = \eta^2 - \xi^2.$
- Canonical form:

$$u_{\eta\eta} = \frac{\eta^2-\xi^2}{4}.$$

- Integrating w.r.t. η: $u_\eta = \frac{1}{4}\left(\frac{\eta^3}{3} - \xi^2\eta\right) + g(\xi).$
- Again: $u(\xi,\eta) = \frac{1}{4}\left(\frac{\eta^4}{12} - \frac{\xi^2\eta^2}{2}\right) + \eta\, g(\xi) + h(\xi)$, with g, h arbitrary.



Example (contd.): General Solution

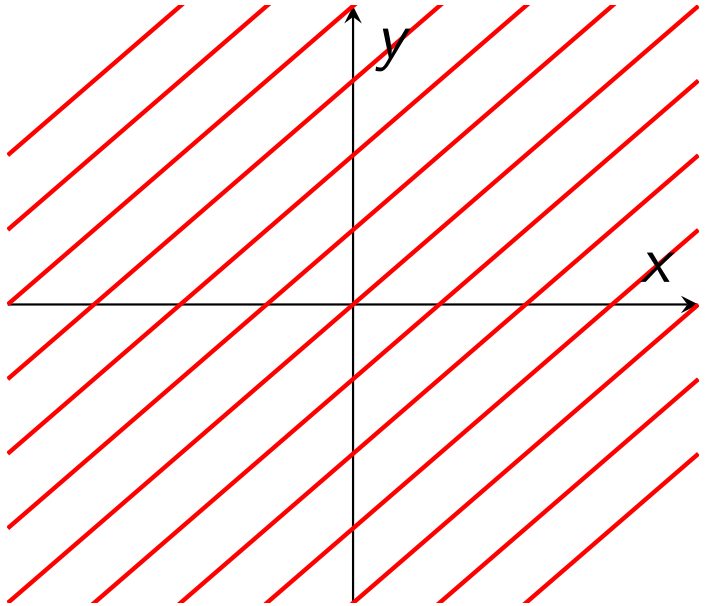
- Return to x, y using $\xi = y - x$, $\eta = x + y$.
- $\dfrac{\eta^4}{48} - \dfrac{\xi^2\eta^2}{8} = \dfrac{1}{48}\left[(x+y)^4 - 6\,(y^2 - x^2)^2\right]$.
- General solution:

$$u(x,y) = \frac{1}{48}\left[(x+y)^4 - 6\,(y^2-x^2)^2\right] + (x+y)\,g(y-x) + h(y-x).$$

- Check the particular part: with $s = x + y$, $w = y - x$, the operator is $(\partial_x + \partial_y)^2$.
- $(\partial_x + \partial_y)s = 2$ and $(\partial_x + \partial_y)w = 0$, so it gives $s^2 - w^2 = 4xy$.
- Only one family of characteristics: the parallel lines $y - x = c_1$.



Picture: Characteristics of $u_{xx} + 2u_{xy} + u_{yy} = 4xy$



- A single family of parallel lines $y - x = c_1$ (the double root $dy/dx = 1$).
- Through each point passes only one characteristic, unlike the hyperbolic case.



Example: $y^2 u_{xx} - 2xy\, u_{xy} + x^2 u_{yy} = \frac{y^2}{x} u_x + \frac{x^2}{y} u_y$

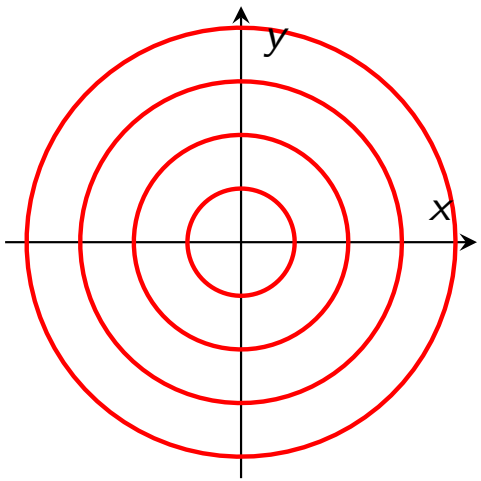
- $(x, y \neq 0.)$ $a = y^2$, $b = -2xy$, $c = x^2$:
  $b^2 - 4ac = 4x^2y^2 - 4x^2y^2 = 0$; parabolic.
- Characteristic equation: $\dfrac{dy}{dx} = \dfrac{b}{2a} = -\dfrac{x}{y} \Rightarrow x\,dx + y\,dy = 0.$
- So $x^2 + y^2 = c_1$: the characteristics are circles about the origin.
- Take $\xi = x^2 + y^2$ and choose $\eta = x^2 - y^2$;
  $J = 2x(-2y) - 2y(2x) = -8xy \neq 0.$
- Then $A = B = 0$ and $C = (2xy + 2xy)^2 = 16x^2y^2.$
- With $d = -\frac{y^2}{x}$, $e = -\frac{x^2}{y}$: $D = 2y^2 + 2x^2 - 2y^2 - 2x^2 = 0.$
- Similarly $E = 2y^2 - 2x^2 - 2y^2 + 2x^2 = 0.$
- Canonical form: $16x^2y^2\, u_{\eta\eta} = 0$, i.e. $u_{\eta\eta} = 0.$



Example (contd.)

- Integrating $u_{\eta\eta} = 0$ twice w.r.t. η: $u = \eta\,\phi(\xi) + \psi(\xi)$.
- General solution:

$$u(x, y) = (x^2 - y^2)\,\phi(x^2 + y^2) + \psi(x^2 + y^2).$$



- The choice of η is not unique: any η with $J \neq 0$ works (the form may then differ).



Elliptic Type: Complex Characteristics

- Suppose $b^2 - 4ac < 0$ in the region (so $a \neq 0$).
- The characteristic equation has complex conjugate roots:

$$\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{b}{2a} \pm i\,\frac{\sqrt{4ac - b^2}}{2a}.$$

- There are no real characteristic curves.
- Let $w(x, y) = \xi(x, y) + i\,\eta(x, y)$ be a complex solution: $w =$ const along

$$a\left(\frac{dy}{dx}\right)^2 - b\,\frac{dy}{dx} + c = 0.$$

- As before, $\dfrac{w_x}{w_y} = -\dfrac{dy}{dx}$, so

$$a\,w_x^2 + b\,w_x w_y + c\,w_y^2 = 0.$$



Elliptic Type: $A = C$ and $B = 0$

- Substitute $w_x = \xi_x + i\eta_x$, $w_y = \xi_y + i\eta_y$:

$$a(\xi_x + i\eta_x)^2 + b(\xi_x + i\eta_x)(\xi_y + i\eta_y) + c(\xi_y + i\eta_y)^2 = 0.$$

- Real part: $(a\xi_x^2 + b\xi_x\xi_y + c\xi_y^2) - (a\eta_x^2 + b\eta_x\eta_y + c\eta_y^2) = 0$, i.e. $A - C = 0$.
- Imaginary part: $2a\xi_x\eta_x + b(\xi_x\eta_y + \xi_y\eta_x) + 2c\xi_y\eta_y = 0$, i.e. $B = 0$.
- So the real variables $\xi = \operatorname{Re} w$, $\eta = \operatorname{Im} w$ give $A = C$, $B = 0$.
- $A \neq 0$, since $B^2 - 4AC = -4A^2 = J^2(b^2 - 4ac) < 0$.
- This also shows $J \neq 0$ automatically.



Canonical Form of an Elliptic PDE

- With $A = C \neq 0$, $B = 0$, equation (7) becomes

$$A\,(u_{\xi\xi} + u_{\eta\eta}) + D\,u_{\xi} + E\,u_{\eta} + F\,u = G.$$

- Dividing by A:

$$u_{\xi\xi} + u_{\eta\eta} = \Phi(\xi, \eta, u, u_{\xi}, u_{\eta}).$$

- This is the canonical form of an elliptic PDE.
- The principal part is the Laplacian in the new variables.
- Procedure: solve the characteristic equation in complex form, take real and imaginary parts.
- The Laplace and Poisson equations are already in canonical form.
- Real characteristics do not exist; complex ones are used only as a device.



Example: $u_{xx} + u_{xy} + u_{yy} = 0$

- $a = b = c = 1$: $b^2 - 4ac = -3 < 0$; elliptic everywhere.
- Characteristic equation: $\dfrac{dy}{dx} = \dfrac{1 \pm i\sqrt{3}}{2}$.
- Integrating: $y - \dfrac{x}{2} \mp i\,\dfrac{\sqrt{3}}{2}\,x = \text{const}$, i.e. $(2y - x) \mp i\sqrt{3}\,x = \text{const}$.
- Take $\xi = 2y - x$, $\eta = \sqrt{3}\,x$;
  $J = (-1)(0) - (2)(\sqrt{3}) = -2\sqrt{3} \neq 0$.
- $u_x = -u_\xi + \sqrt{3}\,u_\eta$, $u_y = 2u_\xi$.
- $u_{xx} = u_{\xi\xi} - 2\sqrt{3}\,u_{\xi\eta} + 3u_{\eta\eta}$, $u_{xy} = -2u_{\xi\xi} + 2\sqrt{3}\,u_{\xi\eta}$,
  $u_{yy} = 4u_{\xi\xi}$.
- Sum: $3u_{\xi\xi} + 3u_{\eta\eta} = 0$.
- Canonical form: $u_{\xi\xi} + u_{\eta\eta} = 0$ (Laplace equation in ξ, η).



Example: $y^2 u_{xx} + u_{yy} = 0, \quad y \neq 0$

- $a = y^2,\ b = 0,\ c = 1$: $\quad b^2 - 4ac = -4y^2 < 0$ for $y \neq 0$; elliptic.
- Characteristic equations:

$$\frac{dy}{dx} = \frac{0 \pm \sqrt{-4y^2}}{2y^2} = \pm \frac{i}{y}.$$

- $y\,dy = \pm i\,dx \Rightarrow \dfrac{y^2}{2} \mp i\,x = \text{const.}$
- Take the real and imaginary parts: $\xi = \dfrac{y^2}{2}, \quad \eta = x.$
- $\xi_x = 0,\ \xi_y = y,\ \eta_x = 1,\ \eta_y = 0; \quad J = -y \neq 0$ for $y \neq 0$.



Example (contd.)

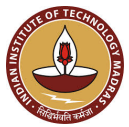
- $u_x = u_\xi\,\xi_x + u_\eta\,\eta_x = u_\eta$, so $u_{xx} = u_{\eta\eta}$.
- $u_y = u_\xi\,\xi_y + u_\eta\,\eta_y = y\,u_\xi$.
- $u_{yy} = u_\xi + y\,(u_{\xi\xi}\,\xi_y + u_{\xi\eta}\,\eta_y) = y^2 u_{\xi\xi} + u_\xi$.
- Substituting in $y^2 u_{xx} + u_{yy} = 0$:

$$y^2 u_{\eta\eta} + y^2 u_{\xi\xi} + u_\xi = 0.$$

- Divide by $y^2 = 2\xi$:

$$u_{\xi\xi} + u_{\eta\eta} = -\frac{u_\xi}{2\xi}.$$

- This is the required canonical form ($\xi > 0$).



Example: Tricomi Equation for $y > 0$

- $y\,u_{xx} + u_{yy} = 0$ is elliptic for $y > 0$.
- Characteristic equations: $\dfrac{dy}{dx} = \dfrac{\pm\sqrt{-4y}}{2y} = \pm\dfrac{i}{\sqrt{y}}$.
- $\sqrt{y}\,dy = \pm i\,dx \Rightarrow \frac{2}{3}y^{3/2} \mp i\,x = \text{const.}$
- Take $\xi = x,\ \ \eta = \frac{2}{3}y^{3/2};\ \ J = \sqrt{y} \neq 0.$
- $u_{xx} = u_{\xi\xi},\ \ u_y = \sqrt{y}\,u_\eta,\ \ u_{yy} = y\,u_{\eta\eta} + \dfrac{1}{2\sqrt{y}}\,u_\eta.$
- Substituting and dividing by y: $u_{\xi\xi} + u_{\eta\eta} + \dfrac{u_\eta}{2y^{3/2}} = 0.$
- Since $y^{3/2} = \frac{3}{2}\eta$, the canonical form is

$$u_{\xi\xi} + u_{\eta\eta} + \frac{1}{3\eta}\,u_\eta = 0.$$



Constant Coefficients: Simplifications

- If a, b, c are constants, the characteristics are straight lines.
- Then ξ, η are linear in x, y, so $\xi_{xx} = \xi_{xy} = \cdots = 0$.
- Hence $D = d\,\xi_x + e\,\xi_y$ and $E = d\,\eta_x + e\,\eta_y$ are constants too.
- Canonical forms with constant D, E, F:
- Hyperbolic: $u_{\xi\eta} + D_1 u_\xi + E_1 u_\eta + F_1 u = G_1$.
- Parabolic: $u_{\eta\eta} + D_1 u_\xi + E_1 u_\eta + F_1 u = G_1$.
- Elliptic: $u_{\xi\xi} + u_{\eta\eta} + D_1 u_\xi + E_1 u_\eta + F_1 u = G_1$.
- The first order terms can be removed further by a change of the dependent variable.



Removing First Order Terms (Hyperbolic)

- Consider $u_{\xi\eta} + D_1 u_\xi + E_1 u_\eta + F_1 u = 0$ (D_1, E_1, F_1 constants).
- Put $u = v\, e^{-E_1\xi - D_1\eta}$ and write $\mathcal{E} = e^{-E_1\xi - D_1\eta}$.
- $u_\xi = (v_\xi - E_1 v)\,\mathcal{E}, \quad u_\eta = (v_\eta - D_1 v)\,\mathcal{E}$.
- $u_{\xi\eta} = (v_{\xi\eta} - D_1 v_\xi - E_1 v_\eta + D_1 E_1 v)\,\mathcal{E}$.
- Substituting and cancelling $\mathcal{E}$:

$$v_{\xi\eta} + (F_1 - D_1 E_1)\, v = 0.$$

- The first order derivatives have disappeared.
- Similar substitutions $u = v\, e^{\lambda\xi + \mu\eta}$ work in the parabolic and elliptic cases.



Example: Removing First Order Terms

- Consider $u_{\xi\eta} + 2u_\xi + 3u_\eta + 5u = 0$; here $D_1 = 2$, $E_1 = 3$, $F_1 = 5$.
- Put $u = v\, e^{-3\xi - 2\eta}$.
- Then $v_{\xi\eta} + (5 - 6)\, v = 0$, i.e. $v_{\xi\eta} - v = 0$.
- Elliptic example: $u_{\xi\xi} + u_{\eta\eta} + 2u_\xi - 4u_\eta = 0$.
- Put $u = v\, e^{\lambda\xi + \mu\eta}$ and choose $\lambda = -1$, $\mu = 2$ (half the coefficients, sign reversed).
- Then $v_{\xi\xi} + v_{\eta\eta} + (\lambda^2 + \mu^2 + 2\lambda - 4\mu)\, v = 0$.
- $\lambda^2 + \mu^2 + 2\lambda - 4\mu = 1 + 4 - 2 - 8 = -5$, so $v_{\xi\xi} + v_{\eta\eta} - 5v = 0$.



Summary of Canonical Forms

- Characteristic equation: $a\left(\frac{dy}{dx}\right)^2 - b\frac{dy}{dx} + c = 0$, i.e. $\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$.

Type	Characteristics	Choice of ξ, η	Canonical form
$b^2 - 4ac > 0$	two real	both characteristic	$u_{\xi\eta} = \Phi$
$b^2 - 4ac = 0$	one real	ξ char., η any	$u_{\eta\eta} = \Phi$
$b^2 - 4ac < 0$	complex pair	$\xi + i\eta$ char.	$u_{\xi\xi} + u_{\eta\eta} = \Phi$

- Second hyperbolic form: $u_{\alpha\alpha} - u_{\beta\beta} = \Phi_1$ with $\alpha = \xi + \eta$, $\beta = \xi - \eta$.
- The type is invariant: $B^2 - 4AC = J^2(b^2 - 4ac)$.
- The type may vary from region to region with the sign of $b^2 - 4ac$.



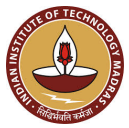
Homework: Canonical Forms (I)

- Find the canonical forms of the following PDEs:
- (1) $4y^2u_{xx} - e^{2x}u_{yy} = 0, \quad y \neq 0.$

  Ans: $\xi = e^x + y^2$, $\eta = e^x - y^2$,
  $$u_{\xi\eta} = \frac{(3\eta - \xi)\, u_\xi + (\eta - 3\xi)\, u_\eta}{4(\xi^2 - \eta^2)}.$$
- (2) $x^2u_{xx} + y^2u_{yy} = 0, \quad x, y \neq 0.$

  Ans: $\xi = \ln y$, $\eta = \ln x$, $\quad u_{\xi\xi} + u_{\eta\eta} = u_\xi + u_\eta.$
- (3) $u_{xx} + x^2u_{yy} = 0, \quad x \neq 0.$

  Ans: $\xi = \frac{1}{2}x^2$, $\eta = y$, $\quad u_{\xi\xi} + u_{\eta\eta} = -\dfrac{u_\xi}{2\xi}.$



Homework: Canonical Forms (II)

- (4) $u_{xx} + 2u_{xy} + u_{yy} = 0.$

  Ans: $\xi = x - y$, choose $\eta = x + y$, $u_{\eta\eta} = 0.$
- (5) $u_{xx} - x^2 u_{yy} = 0, \quad x \neq 0.$

  Ans: $\xi = y + \frac{x^2}{2}$, $\eta = y - \frac{x^2}{2}$, $u_{\xi\eta} = \dfrac{u_\xi - u_\eta}{4(\xi - \eta)}.$
- (6) $u_{xx} + 4u_{xy} + 3u_{yy} = 0$; also find the general solution.
- (7) $u_{xx} + 2u_{xy} + 5u_{yy} + u_x = 0.$
- (8) $x\, u_{xx} + u_{yy} = x^2, \quad x > 0.$
- (9) $u_{xx} + y\, u_{yy} + \frac{1}{2} u_y = 0, \quad y < 0.$



Homework: Mixed Problems

- (10) Find the regions of hyperbolicity, parabolicity and ellipticity of $y\,u_{xx} + 2u_{xy} + u_{yy} = u\,u_x + u_y^2$ and sketch them.
- (11) Reduce $u_{xx} - 2\sin x\,u_{xy} - \cos^2 x\,u_{yy} - \cos x\,u_y = 0$ to canonical form.
- (12) Show that $u_{xx} - 2u_{xy} + u_{yy} = 0$ has general solution $u = \phi(x+y) + x\,\psi(x+y)$.
- (13) Reduce $u_{xx} + 2u_{xy} + 17u_{yy} = 0$ to $u_{\xi\xi} + u_{\eta\eta} = 0$.
- (14) Verify $B^2 - 4AC = J^2(b^2 - 4ac)$ by direct expansion.
- (15) Remove the first order terms from $u_{\xi\eta} - 4u_\xi + u_\eta + 3u = 0$.
- In each problem draw the characteristic curves where they exist.

